Radians

1 Calculate the following trigonometric ratios to two decimal places:

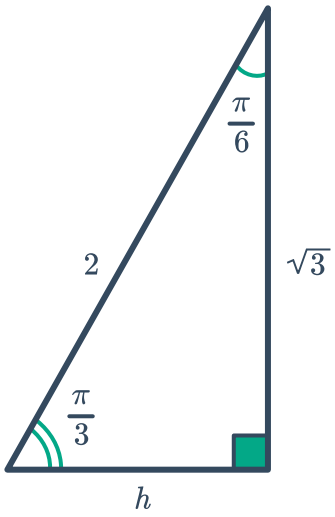
a $\sin\dfrac{35\pi}{16}$ **b** $\cos 6.87$ **c** $\sin 7.26$ **d** $\tan 7.26$

e $\tan\left(\dfrac{-3\pi}{7}\right)$ **f** $\cos\dfrac{2\pi}{3}$ **g** $\sin\left(-\dfrac{4\pi}{3}\right)$ **h** $\cos\dfrac{4\pi}{5}$

i $\sin\left(-\dfrac{4\pi}{5}\right)$

Exact values

2 Consider the following diagram:



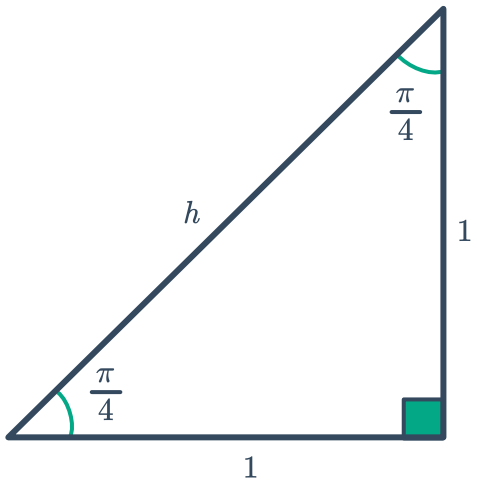
a Find the length of side h.

b Hence, state the exact value of:

i $\sin\dfrac{\pi}{3}$ **ii** $\sin\dfrac{\pi}{6}$

iii $\tan\dfrac{\pi}{3}$ **iv** $\cos\dfrac{\pi}{3}$

3 Consider the following diagram:



a Find the length of the hypotenuse, h.

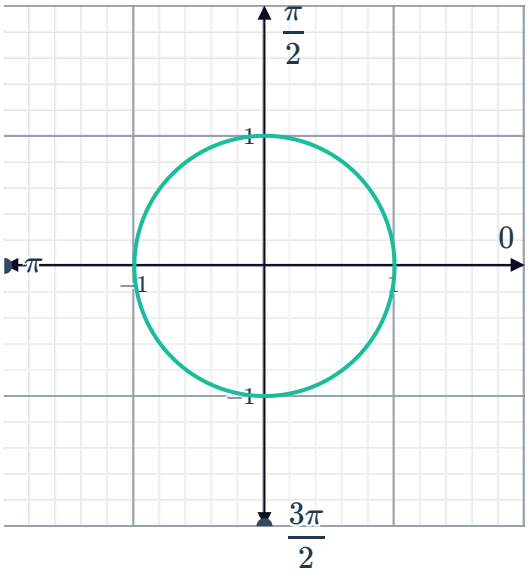
b Hence, state the exact value of:

i $\sin\dfrac{\pi}{4}$ **ii** $\cos\dfrac{\pi}{4}$

iii $\tan\dfrac{\pi}{4}$

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4 Consider the unit circle diagram and state the exact value of the following trigonometric ratios:



a $\sin\frac{\pi}{2}$ **b** $\cos\frac{3\pi}{2}$

c $\tan\pi$ **d** $\cos 0$

e $\sin(-2\pi)$

5 Find the exact value of the following:

a $\sin\frac{\pi}{3}+\cos\frac{\pi}{3}$ **b** $\sin\frac{\pi}{6}\cos\frac{\pi}{4}$

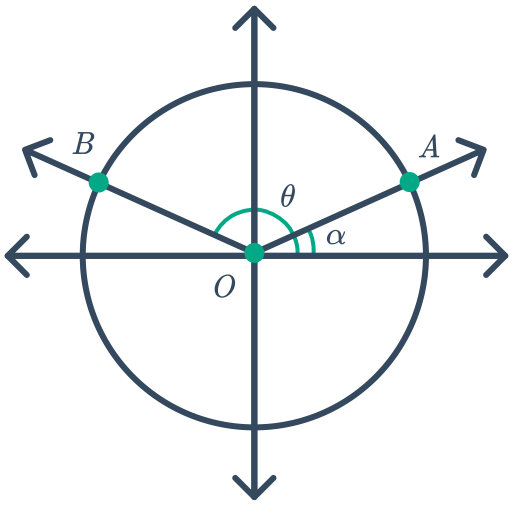
c $\dfrac{\sin\frac{\pi}{3}}{\cos\frac{\pi}{6}}$ **d** $\sin\frac{\pi}{4}\cos\frac{\pi}{6}+\tan\frac{\pi}{4}$

e $\sin^2\left(\frac{\pi}{6}\right)-\cos^2\left(\frac{\pi}{3}\right)$ **f** $2\sin^2\left(\frac{\pi}{2}\right)+3\cos^2\left(\frac{\pi}{2}\right)$

Exact values from reference angles

6 Consider the unit circle shown, where points A and B have the same y-coordinates.

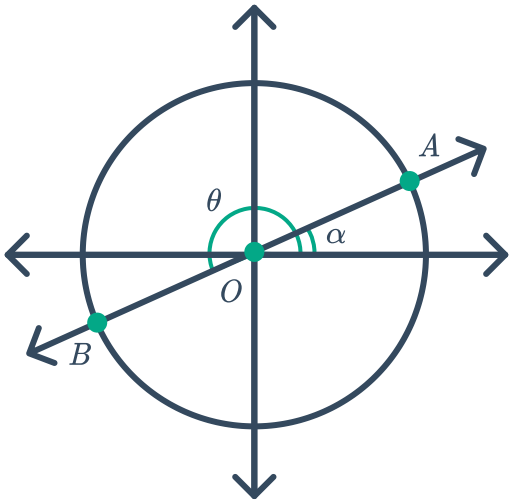
Suppose that $\theta=\dfrac{10\pi}{11}$. State the size of the reference angle, α.



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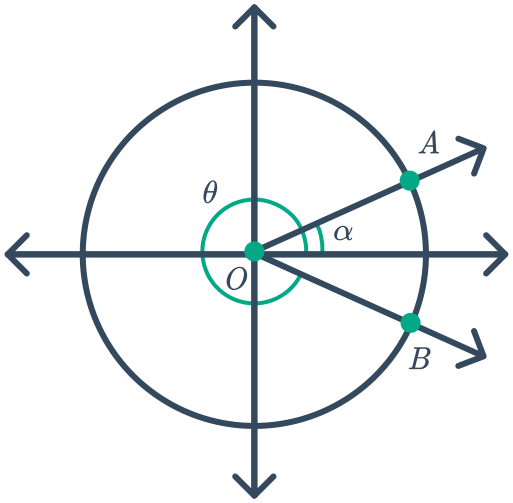
7 Consider the unit circle shown, where the line through A and B passes through the origin, O.

Suppose that $\theta = \frac{8\pi}{7}$. State the size of the reference angle, α.



8 Consider the unit circle shown, where the points A and B have the same x-coordinate.

Suppose that $\theta = \frac{9\pi}{5}$. State the size of the reference angle, α.



9 Find the exact value of the following:

a $\sin\frac{5\pi}{6}$ **b** $\tan\frac{3\pi}{4}$ **c** $\sin\frac{7\pi}{6}$ **d** $\cos\frac{7\pi}{6}$

e $\sin\frac{5\pi}{3}$ **f** $\cos\frac{5\pi}{3}$ **g** $\cos 4\pi$ **h** $\tan 9\pi$

i $\sin\frac{5\pi}{2}$ **j** $\cos\frac{7\pi}{2}$ **k** $\cos\frac{3\pi}{4}$ **l** $\sin\frac{5\pi}{4}$

m $\cos\frac{5\pi}{4}$ **n** $\tan\frac{5\pi}{4}$ **o** $\tan\frac{7\pi}{6}$ **p** $\tan\frac{11\pi}{6}$

10 Find the exact value of the following:

a $\sin\left(-\frac{17\pi}{6}\right)$ **b** $\cos\left(-\frac{17\pi}{6}\right)$ **c** $\cos\left(-\frac{4\pi}{3}\right)$ **d** $\tan\left(-\frac{17\pi}{6}\right)$

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11 Find the exact value of the following:

a $\dfrac{\left(\sin\dfrac{2\pi}{3}\right)\left(\cos\dfrac{2\pi}{3}\right)\left(\tan\dfrac{3\pi}{4}\right)}{\tan\left(-\dfrac{\pi}{4}\right)}$

b $\dfrac{\sin\dfrac{2\pi}{3}+\cos\dfrac{5\pi}{6}-\tan\dfrac{7\pi}{4}}{\cos\dfrac{4\pi}{3}}$

Trigonometric functions

12 Consider the equation $y=\sin x$.

a Complete the table with values in exact form:

x	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{3\pi}{2}$	$\frac{11\pi}{6}$	2π
$\sin x$									

b Sketch a graph for $y=\sin x$ on the domain $-2\pi\leq 0\leq 2\pi$.

c State the value of $\sin(-2\pi)$.

d State the sign of $\sin\left(\dfrac{-\pi}{12}\right)$.

e State the sign of $\sin\dfrac{13\pi}{12}$.

f Which quadrant of a unit circle does an angle with measure $\dfrac{13\pi}{12}$ lie in?

13 Consider the equation $y=\cos x$.

a Complete the table with values in exact form:

x	0	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	2π
$\cos x$									

b Sketch a graph for $y=\cos x$ on the domain $-2\pi\leq 0\leq 2\pi$.

c State the value of $\cos\pi$.

d State the sign of $\cos\left(\dfrac{-\pi}{4}\right)$.

e State the sign of $\cos\dfrac{11\pi}{6}$.

f Which quadrant of a unit circle does an angle with measure $\dfrac{11\pi}{6}$ lie in?

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14 Consider the equation $y = \tan x$.

a Complete the table with values in exact form:

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\tan x$									

b Sketch the graph of $y = \tan x$ on the domain $-2\pi \leq 0 \leq 2\pi$.

c Graph the line $y = 1$ on the same coordinate plane.

d Hence, state the exact solutions to the equation $\tan x = 1$ over this domain.

e State the value of $\tan(-2\pi)$.

f State the sign of $\tan\left(\frac{-\pi}{6}\right)$.

g State the sign of $\tan\frac{9\pi}{5}$.

h Which quadrant of a unit circle does an angle with measure $\frac{9\pi}{5}$ lie in?

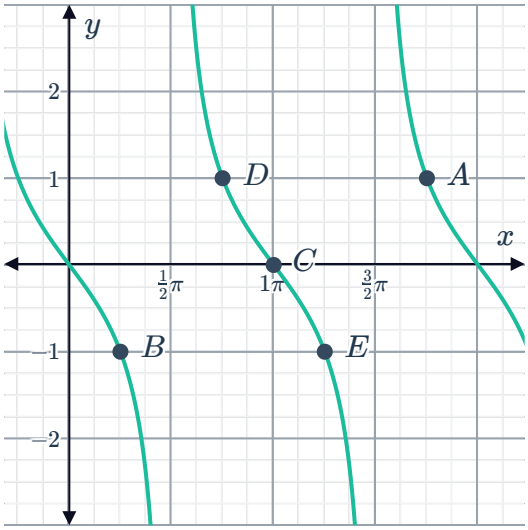
15 Consider the function $y = 5\sin x$.

a Sketch the graph of the given function over the domain $[-2\pi, 2\pi]$.

b Graph the line $y = x$ on the same coordinate plane.

c Hence, state the number of solutions to the equation $5\sin x = x$.

16 Consider the graph of $y = -\tan x$ and the plotted points A, B, C, D and E shown:
At which point is the graph of $y = -\tan x$ equal to zero?



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